# Anonymization for Patient data @ Neural Wave Hackathon

**Introduction**

Patient data from hospitals is of high confidentiality and hence using it with cloud-based LLMs requires utmost precaution. Therefore, it is necessary to anonymize the data beforehand. However, anonymizing comes at the cost of losing information in the data, which reduces its utility. The task of this hackathon challenge is to develop anonymization methods which allow for as much data utility as possible while making sensitive information unidentifiable.

**Data**

The file patient_doctor_encounters.jsonl (Source: Yuan, Moy, et al. "Toward reliable clinical coding with language models: verification and lightweight adaptation." *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track*. 2025. Repository, License, entries in dataset unmodified) contains recorded dialogues of doctors with their patients as well as medical notes by the doctor. These encounters contain sensitive private and medical information. Each encounter has one or more diagnoses associated with it. They are specified by the fields “code” (referring to ICD-10 code, a standard for medical diagnoses) and “description” under “icd10_codes”. File eval_patient_doctor_encounters.jsonl contains the evaluation split with recorded dialogues of the same fashion and can be used for your own evaluation during the hackathon. All information in the data is available during inference.

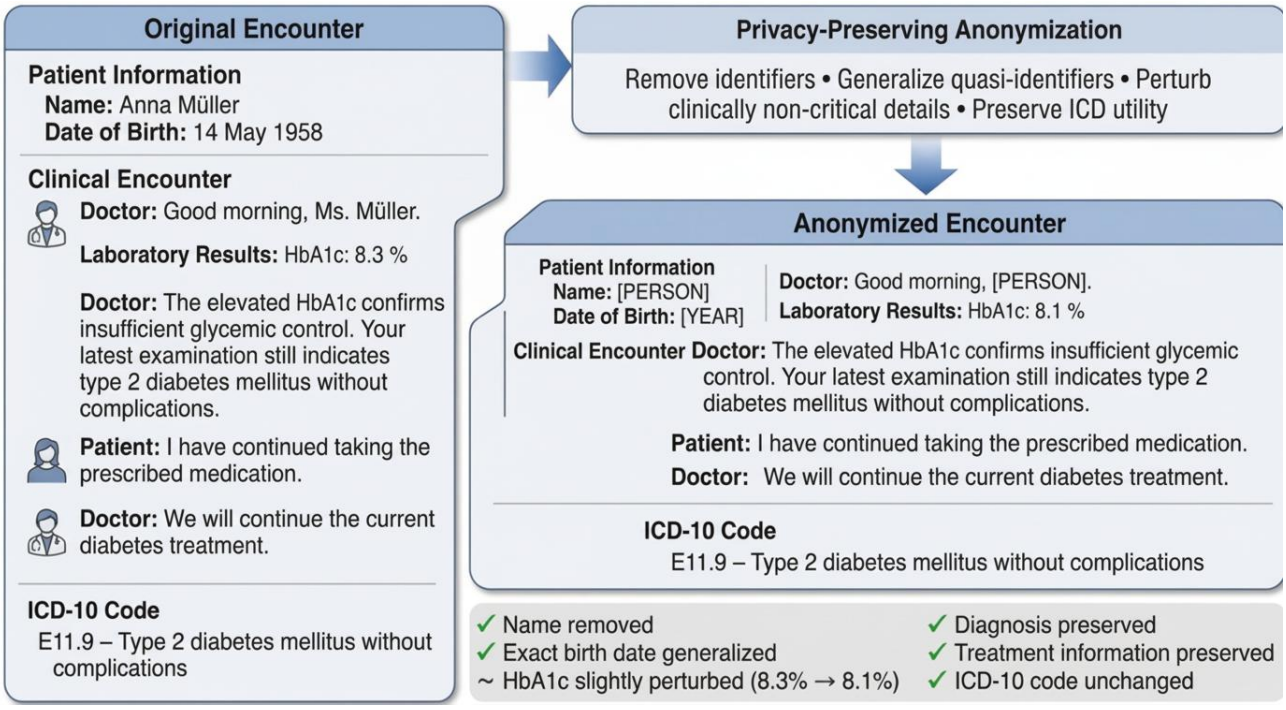
| **Field** | **Description** | **Available for anonymization?** |
|---|---|---|
| dialogue | Conversation between patient and doctor | Yes |
| note | Notes written by physician | Yes |
| icd10_codes.code | ICD-10 diagnosis code | Must be preserved |
| icd10_codes.description | Diagnosis description | Must be preserved |

**Task**

Develop anonymization techniques using the provided dataset for private and clinical information. This involves identifying sensitive clinical information and critical privacy-related (e.g., names, date of birth,...) information. Differentiate between the two (e.g., Alzheimer and Parkinson can be names for diagnoses and/or person’s last names). Possible directions for a start (others are possible too and welcome):

i. What are suitable anonymization techniques for each case to be anonymized? How can they be combined?
ii. Think about how to reduce the potential of privacy attacks (e.g., based on rare words, non-medical context, quasi-identifiers, temporal information,... )

Example:




The developed solution needs to obey the following constraints:

i. Anonymization needs to run on-prem (can't use any cloud-hosted model, max. 4 GB VRAM). This only applies during evaluation, for development additional computational resources may be used.
ii. ICD-10 codes need to be preserved as much as possible. If it doesn't affect the ICD-10 code, diagnosis-relevant information should be anonymized if needed for privacy concerns

**Bonus Tasks (Optional)**

Participants may optionally tackle one or more bonus tasks. These tasks are not required for a valid submission, however they will be considered in the final ranking.

- **Adversarial Privacy Attacks:** Demonstrate the robustness of the anonymization pipeline by attempting to recover removed information using red-teaming or adversarial attacks.
- **Explainability:** Provide explanations for anonymization decisions, e.g., why certain information was removed, generalized, perturbed, or retained.
- **Efficiency Optimization:** Minimize inference latency, memory consumption, or computational cost while maintaining anonymization performance.
- **Without ICD-10 codes and their description available during inference:** Relevant information from dialogues and note for ICD-10 code classification should still be preserved, but the code is unknown.
- **Robustness to other types of input data**: Demonstrate the anonymization approach on previously unseen writing styles, note formats or other medical contexts (e.g., hospital). Data may be synthetically generated.

## Evaluation

Developed solutions are compared based on how much they protect patient privacy, the usefulness of the remaining medical information as well as their originality and scalability (each one having equal weight). Metrics are computed using a non-public dataset with the same structure.

i. Evaluation for privacy protection: based on a fixed evaluation pipeline provided by the organizers.
   It compares anonymized with non-anonymized data on how much private information was leaked
   *Used metrics:*
   a. Proportion of removed privacy-related information
ii. Evaluation of preserving medical information: based on a fixed evaluation pipeline provided by the organizers.
   It compares anonymized with non-anonymized data on how much ICD-10-code relevant information was kept.
   *Used metrics:*
   a. F1-Score ICD-10 code prediction
   b. Medical concept recall
iii. Assessment of originality and scalability (evaluated by organizers and jury)
iv. Bonus tasks
   a. Adversarial Privacy Attacks:
      i. Sophistication of the attack
      ii. Proportion of recovered information
   b. Explanation:
      i. Percentage of anonymization decisions having an explanation
      ii. Correctness and quality of explanations (evaluated by jury)
   c. Efficiency Optimization:
      i. Runtime
      ii. Peak RAM
      iii. Peak VRAM
   d. Without ICD-10 codes available during inference
      i. Same metrics as main task
   e. Robustness:
      i. Complexity of additional data
      ii. Same metrics as main task

There's a trade-off between the first two criteria. Hence, try out different approaches and compare them based on these.

Each team can request one mid-term evaluation by sending their anonymized data set to Jonas.Bergdolt@bosch.com. You will receive the value of the metrics back.

## Expected deliverables

i. Anonymized dataset for evaluation as .jsonl
ii. Repository containing:
    a. Source code of your anonymization tool
    b. README.md on how to use your tool